

Review

Passively stable dissemination of ultrastable optical frequency via a noisy field fiber network

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ABSTRACT

Optical fibers have been recognized as one of the most promising host material for coherent optical frequency transfer over thousands of kilometers. In the pioneering work, the active phase noise cancellation (ANC) technique has been widely used for suppressing the fiber phase noise introduced by the environmental perturbations, in which an ideal phase detector with high resolution and unlimited detection range is needed to extract the fiber phase noise, in particular for noisy fiber links. We demonstrate the passive phase noise cancellation (PNC) technique without the need of phase detector could be preferable for noisy fiber links. To avoid the effect of the radio frequency (RF) from the time base at the local site in the conventional active or passive phase noise cancellation techniques, here we introduce a fiber-pigtailed acousto-optic modulator (AOM) with two diffraction order outputs (0 and -1 order) with properly allocating the AOM-driving frequencies allowing to cancel the time base effect. Using this technique, we demonstrate transfer of coherent light through a 260 km noisy urban fiber link. The results show the effect of the RF reference can be successfully removed. After being passively compensated, we demonstrate a fractional frequency instability of 4.9×10^{-14} at the integration time of 1 s and scales down to 10^{-20} level at 10,000 s in terms of modified Allan deviation over the 260 km noisy urban fiber link. The frequency uncertainty of the retrieved light after transferring through this noise-compensated fiber link relative to that of the input light achieves $(0.41 \pm 4.7) \times 10^{-18}$. The proposed technique opens a way to a broad distribution of an ultrastable frequency reference with high coherence without any effects coming from the RF reference and enables a wide range of applications beyond metrology over fiber networks.

1. Introduction

Optical frequency timekeeping with trapped atoms or ions [1–3] is progressively becoming ideal tools for precision measurements and fundamental physics tests, such as general relativity [4,5], temporal variation of the fundamental constant [6], searching for dark matter [7], chronometric geodesy [8], and gravitational waves [9–11]. Unfortunately, these clocks are typically cumbersome and expensive and only available at national metrology institutes and several universities, causing a strong motivation to develop a robust and cost-effective system for comparing and distributing these ultraprecise frequency signals. Among different frequency transfer methods, optical fiber links have been widely demonstrated as one of the most powerful tools for

the dissemination and comparison of current state-of-the-art optical clocks with the instability and uncertainties at a few 10^{-20} level [12–14], which outperforms the traditional satellite-based techniques by at least three orders in terms of transfer instability [15]. However, the temporal optical length variations of the fiber link, coming from mechanical and thermal perturbations along the fiber link, will imprint optical phase noise onto the transmitted optical signal and ultimately deteriorate the optical frequency transfer instability and uncertainty. One of the most pioneering technique, namely active phase noise cancellation (ANC), has been widely adopted for mitigating this fiber-induced Doppler noise, where the phase noise of the fiber link is

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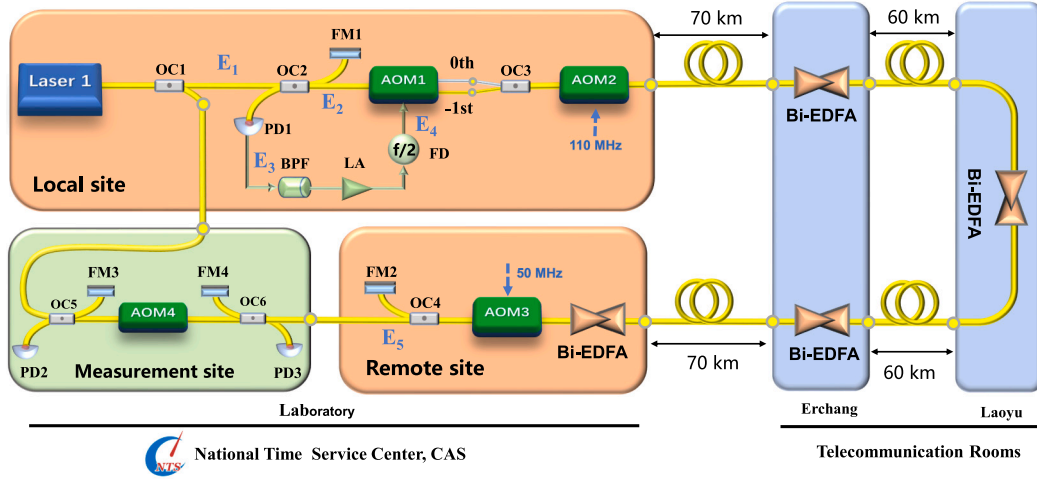


Fig. 1. Schematic of optical frequency transfer via a 260 km urban fiber link by adopting the passive phase noise cancellation technique without the effect of the time base. AOM: acousto-optic modulator, FM: Faraday mirror, Bi-EDFA: bi-directional erbium-doped fiber amplifier, OC: optical coupler, FD: frequency divider, PD: photo-detector, BPF: band-pass filter, LA: logarithmic amplifier.

detected by comparing the local signal with the round-trip one and be used to implement a feedback control onto the phase compensator [12]. Nevertheless, there exist obvious and inherent disadvantages for this ANC technique that it requires the complicated phase noise detection unit and the sophisticated servo controller. The overall performance of the phase noise rejection capability is strongly dependent on the proper setting of the loop parameters. At the same time, an ideal phase detector with high resolution and unlimited detection range is needed to extract the fiber phase noise. One common way to extend the phase detection range is to adopt an analog phase detector with an assistance of a high frequency dividing ratio [16] or a digital phase detector with the proper phase unwrap algorithm [17]. Consequently, a complicated phase noise cancellation circuit has to be used in a noisy fiber link with a large phase noise power spectral density (PSD) per-unit-length such as $4 \text{ rad}^2/\text{Hz}/\text{km}$ (USA) [18], $0.5 \text{ rad}^2/\text{Hz}/\text{km}$ (Germany) [19], $3 \text{ rad}^2/\text{Hz}/\text{km}$ (France) [16,20], $20 \text{ rad}^2/\text{Hz}/\text{km}$ (Italy) [14], $20 \text{ rad}^2/\text{Hz}/\text{km}$ (Japan) [21] at the 1 Hz offset frequency. The measured phase noise is even larger in this study, the phase noise PSD per-unit-length at the 1 Hz is about $38.5 \text{ rad}^2/\text{Hz}/\text{km}$ for a 260 km field fiber link and the integrated phase from 0.1 Hz to 100 kHz in the frequency domain are approaching to 240 rad (see the following text). For an analog phase detector, a frequency divider with a ratio of more than 240 is needed to extract the phase noise introduced by the fiber link. To secure the enough phase detection range caused by the fluctuations of the fiber link phase noise, the frequency divider with the larger frequency divider is needed. This could still not be enough for the phase noise of the fiber link fluctuated by more than one order of magnitude, resulting in the appearance of the cycle slips [16]. Moreover, before adopting the frequency divider with the high dividing ratio, the beatnote signal at the local site typically needs to be up-converted as implemented in [16]. This process will introduce additional phase noise coming from the frequency mixer and amplifier stages.

To surmount the aforementioned difficulties, Hu. et al. first proposed a passive phase noise cancellation (PNC) technique for optical frequency transfer by detecting the phase noise with a pilot optical signal and pre-compensating the same amount phase noise on the other forward optical signal, resulting in the phase noise compensated light automatically received at the remote site [22,23]. They demonstrated a proof-of-principle optical frequency transmission experiment via a 145 km spooled fiber link with an instability of 2×10^{-19} at 10,000 s in terms of overlapping Allan deviation (OADEV) [22]. Although the PNC technique has been demonstrated with the spooled fiber link [22], the capability for the phase noise rejection in the field-deployed fiber

links needs to be experimentally examined as widely done by the ANC technique [13,14,16,18,24–27].

Neither the active nor the passive phase compensation schemes can avoid the influence of the time base at the local site, because for a practical application, the radio frequency (RF) signals synthesized from their time bases are independent of the optical frequency reference at the local site. By taking the effect of the time base into consideration, the limitation of the optical frequency transfer instability σ_y can be expressed as.

$$\sigma_y \approx \omega_{RF} * \sigma_{RF} / \omega_{laser} \quad (1)$$

where ω_{laser} stands for the transmitted optical frequency. Consider a case that a customized OCXO is used as the RF time base with $\omega_{RF} = 2\pi \times 10 \text{ MHz}$ and instability of 7.2×10^{-11} at 10,000 s, it will constrain the achievable transfer instability σ_y of an optical frequency at $\omega_{laser} = 2\pi \times 193 \text{ THz}$ to a minimum of 3.7×10^{-18} at 10,000 s, illustrating that the long-term instability will become the limitation. One way to solve this issue is to discipline the RF reference to the GPS signal [28], but this will increase the complexity of the system and the system performance is strongly dependent on the GPS performance. Therefore, one intriguing question is how to distribute the optical signal without the effect of the time base. In a pioneering work, Wu et al. demonstrated the effect of the time base can be effectively removed by cascading one more AOM just after the remote AOM for optical frequency transfer over a branching fiber network [29]. However, adding one more AOM, in particular a fiber-pigtailed one, will introduce the significant outside path, resulting in the increase of the outside phase noise as discussed in [30–32]. Additionally, the scheme still could not get rid of the effect of the time base at the local site.

In this paper, a fiber-pigtailed AOM with two diffraction order outputs (0 and -1 order) is employed as the phase noise compensated device, the retrieved optical frequency signal is independent of the time base at the local site. Furthermore, we demonstrate the PNC technique has unlimited phase noise detection range and resolution and could be preferred for noisy fiber links. We for the first time demonstrate the proposed technique by coherently transferring an ultra-stable optical frequency signal via a 260 km noisy urban fiber link.

The article is organized as follows. We illustrate the concept of coherent optical phase dissemination with the PNC technique without the effect of the time-base in Section 2. We present the experimental set-up and experimental results in Section 3. Finally, we give a conclusion in Section 4.

2. Principle

2.1. Concept of canceling the time base effect in an optical frequency transfer network

Fig. 1 illustrates a schematic diagram of optical frequency transfer via a fiber with the PNC technique. Part of this light source is extracted as the local reference, while the remaining part is injected into the AOM1 with two diffracted outputs (0 and -1 order). The laser outputs from the two fiber-pigtails are further combined and coupled to the AOM2 working at the angular frequency of ω_l (upshifted mode, $+1$ order) and then propagated to the remote site via the fiber link. At the remote site, another AOM3 (upshifted mode, $+1$ order) working at the angular frequency of ω_r is used to discriminate the desired optical signal from the stray reflections by the fiber connectors and splices. The light is retro-reflected back to the local site and the phase noise of the fiber link is recovered by the photodetector 1 (PD1) at the local site. Subsequently, the frequency of the detected signal is divided by a factor of 2 and then drives the AOM1 with the -1 st diffraction order. At the remote site, the phase noise is automatically canceled without the effect of the time base. In brief, the transmitted light output from the 0th order of AOM1 is used to obtain the round-trip signal which contains the information of the fiber-induced phase noise, while the output from the -1 st order is used for the optical phase noise compensation.

The detailed principle of optical frequency transfer is as follows. The initial optical frequency signal can be denoted as,

$$E_1 \propto \cos[\omega_{laser}t + \phi_{laser}], \quad (2)$$

where ω_{laser} and ϕ_{laser} respectively represent the angular frequency and initial phase value of the optical signal. Assuming the optical fiber phase noise induced by environmental perturbations in either direction is equal, the electric field of the light which is reflected by the Faraday mirror 2 (FM2) and transferred back to the local site can be written as,

$$E_2 \propto \cos[(\omega_{laser} + 2\omega_l + 2\omega_r)t + \phi_{laser} + 2(\phi_l + \phi_f + \phi_r)], \quad (3)$$

where ϕ_l and ϕ_r , respectively, represent the initial phases of the driving frequencies of the AOM2 and AOM3, and the ϕ_f is the optical fiber induced phase noise. At the local site, the heterodyne beat-note between E_1 and E_2 is detected by the PD1. We select the electrical signal with the angular frequency of $2(\omega_l + \omega_r)$, which contains the fiber phase noise information, by using a narrow bandpass RF filter, resulting in,

$$E_3 \propto \cos[2(\omega_l + \omega_r)t + 2(\phi_l + \phi_f + \phi_r)]. \quad (4)$$

Afterwards, the angular frequency of E_3 is divided by a factor of 2, obtaining,

$$E_4 \propto \cos[(\omega_l + \omega_r)t + (\phi_l + \phi_f + \phi_r)]. \quad (5)$$

To passively compensate the phase noise along the fiber link, we feed the amplified electrical signal E_4 to the AOM1. The phase noise compensated light output at remote site can be expressed as,

$$E_5 \propto \cos[\omega_{laser}t + \phi_{laser}]. \quad (6)$$

As can be seen, the frequency and phase of the retrieved optical signal are independent of the RF driver signals for AOM2 and AOM3. Consequently, the RF driver signal source with the high frequency instability in our scheme is no longer needed, which will simplify the system and make it cost-effective. There is no frequency shift between the retrieved optical signal and original transmission optical signal, the AOM4 and two Michelson-type fiber interferometers are introduced to construct a two-way optical frequency comparison setup as described in [32,33]. The frequency values of two end's beat notes ω_{PD2} and ω_{PD3} are simultaneously recorded by using a dead-time free frequency counter, and the frequency value of $(\omega_{PD2} - \omega_{PD3})/2$ is used for calculating the transfer instability.

2.2. Delay effect on coherent optical frequency transfer in the PNC scheme

It is important to emphasize that the above description does not take the fiber propagation delay into the account. As demonstrated by Williams et al., the phase noise rejection capability is limited by the propagation delay [18]. Here, we for the first time perform a theoretical analysis of the delay effect in this proposed PNC scheme at the local site, following the approach described in [34]. Assuming the uniform spatial distribution of fiber-induced phase noise, the $\delta\varphi(z, t)$ represents the phase noise at time t and position z , and the uncompensated phase noise of the delivered optical signal at the remote link can be written as

$$\varphi_r(t) = \int_0^L \delta\varphi(z, t - \tau + \frac{z}{c_n}) dz. \quad (7)$$

Where L is the link length, c_n is the light speed in the fiber, and $\tau = L/c_n$ is the single-pass propagation delay. Meanwhile, the accumulated phase noise $\varphi_{rt}(t)$ after a round-trip at the local site can be expressed as

$$\varphi_{rt}(t) = \int_0^L [\delta\varphi(z, t - 2\tau + \frac{z}{c_n}) + \delta\varphi(z, t - \frac{z}{c_n})] dz. \quad (8)$$

which is detect by PD1 and further divided by a factor of 2, resulting in a forward error signal $\varphi_{error}(t)$ at the remote link. By adopting a first-order Taylor expansion, the error signal has an expression of,

$$\begin{aligned} \varphi_{error}(t) &= \frac{1}{2} \varphi_{rt}(t - \tau) \\ &= \frac{1}{2} \int_0^L [\delta\varphi(z, t - 3\tau + \frac{z}{c_n}) + \delta\varphi(z, t - \tau - \frac{z}{c_n})] dz \\ &\approx \int_0^L \delta\varphi(z, t - 2\tau) dz \end{aligned} \quad (9)$$

The remaining phase noise $\varphi_{r,comp}(t)$ at the link termination can be derived by subtracting the obtained error phase noise $\varphi_{error}(t)$ from the uncompensated single-pass phase noise $\varphi_r(t)$.

$$\begin{aligned} \varphi_{r,comp}(t) &= \varphi_r(t) - \varphi_{error}(t) \\ &\approx \int_0^L [\delta\varphi(z, t - \tau + \frac{z}{c_n}) - \delta\varphi(z, t - 2\tau)] dz \\ &\approx \int_0^L (\tau + \frac{z}{c_n}) \frac{\partial}{\partial t} \delta\varphi(z, t) dz \\ &= \int_0^L h(z, t) * \delta\varphi(z, t) dz \end{aligned} \quad (10)$$

Following the analysis method proposed in [34], the impulse response $h(z, t) = (\tau + \frac{z}{c_n})\delta(t)$ and the transfer function $|H(z, f)|^2 = (2\pi f(\tau + \frac{z}{c_n}))^2$ in our case. It can be clearly seen that, there exist an additional time delay τ compared with that in [34]. Consequently, the residual phase noise PSD, $S_{r,comp}(f)$, at the remote site in terms of the single-pass free-running phase noise PSD, $S_{fiber}(f)$, and the single-pass fiber link L propagation delay, τ , can be expressed as [22,31,34],

$$\begin{aligned} S_{r,comp}(f) &= \int_0^L |H(z, f)|^2 * \frac{S_{fiber}(f)}{L} dz \\ &= \int_0^L (2\pi f)^2 * (\frac{z}{c_n} + \tau)^2 * \frac{S_{fiber}(f)}{L} dz \\ &= \frac{7}{3} (2\pi f \tau)^2 S_{fiber}(f) \end{aligned} \quad (11)$$

When the residual phase noise PSD $S_{r,comp}(f)$ is equal to the free-running phase noise PSD $S_{fiber}(f)$, that is $7/3 * (2\pi f \tau)^2 = 1$. The noise-suppressed bandwidth of the PNC technique, f_{PBW} can be obtained,

$$f_{PBW} = \sqrt{\frac{12}{7\pi^2}} * \frac{1}{4\tau} \approx \frac{1}{\sqrt{6}} f_{ABW} \quad (12)$$

The result indicates that, the phase noise rejection capability is proportional to τ^2 both in the PNC and conventional ANC configurations [18,31,34]. Although the residual phase noise PSD and the

corresponding delay-limited bandwidth in this proposed PNC scheme turn out to be a factor of 7 and $\sqrt{6}$ worse than that in the conventional ANC scheme. On the other hand, as it has no need of the active servo loop, the unwanted phase noise introduced by the effect of servo bumps inherent in the ANC scheme can be automatically avoided, which will significantly increase with the fiber distance, especially along the noisy fiber link [16,35]. Additionally, this proposed PNC scheme can simplify the electrical structure of the system.

3. Experimental setup and results

3.1. Experimental setup

We have demonstrated the proposed technique by using the configuration as shown in Fig. 1. The interferometer is built with fiber optics. A cw laser source operating at a center wavelength of 1550.12 nm, whose linewidth is reduced to approximately 1 Hz by locking to a high-finesse Fabry–Perot cavity at a frequency near 193 THz is adopted as an optical reference [36]. The optical fiber is composed of a pair of 130 km fiber link connecting the NTSC (National Time Service Center) at Lintong and two communication rooms of CTCC (China Telecom), with a total optical attenuation of 75 dB. To compensate the optical loss, four bi-directional erbium-doped fiber amplifiers (Bi-EDFAs) are installed along this fiber link. As shown in Fig. 1, the RF driver power output from the logarithmic amplifier (LA) is set to guarantee an efficiency of 50% for the -1 order diffraction and 15% for the 0 order diffraction, which can ensure a continuous phase noise measurement. In this experiment, we set $\omega_l = 2\pi \times 110$ MHz and $\omega_r = 2\pi \times 50$ MHz. In order to conveniently evaluate the performance of this coherent frequency transmission system, a two-way optical frequency comparison setup is adopted as described in [32,33]. We use non-averaging Π -type frequency counters to record the beating frequency. Additionally, to measure the phase noise of optical frequency transfer, a commercial phase noise analyzer (Rohde & Schwarz, FSWP) is used.

3.2. Residual phase noise PSDs at the remote site

Fig. 2(a) shows the measured phase noise PSDs of the free-running (blue curve) and stabilized (red curve) 260 km field fiber link. The frequency resolution bandwidth is set as 1 mHz, resulting in 11 min for a single measurement. The free-running phase noise PSD falls off approximately as $10000/f^2$ between 0.1 Hz and 10 Hz, the f stands for the Fourier frequency. The results shows that the long-haul fiber link performs a higher level of phase noise PSD per-unit-length at the 1 Hz (about $38.5 \text{ rad}^2/\text{Hz}/\text{km}$), which is much larger than the fiber noise observed in previous reports [14,16,18–21,35]. Furthermore, the phase noise PSD of this free-running fiber link exists obvious broad bumps around 1.4 Hz and 15 Hz. The peak variations around these two frequencies appear to be related to human activities, we can reasonably attribute this feature to infra-sound perturbations arising from urban traffic [37]. To further characterize the free-running link with the different times, we measured phase noise PSDs of the 260 km free-running fiber for a whole day. The measured total 138 curves are shown in Fig. 2(b). We can observe, the phase noise at the frequency range ($f < 1$ Hz) will fluctuate more than one order of magnitude. We attribute the high phase noise to the fact that this 260 km fiber link is laying underground along a noisy urban area. The measured passively stabilized phase noise PSD (red curve) also shown in Fig. 2(a), which is in good agreement with the PNC theoretical limit (gray dash line) as indicated in Eq. (11). More importantly, the strong servo bumps which existing in the ANC technique is disappeared, resulting in the reduction of the integrate timing jitter of the transferred light [22,23,31].

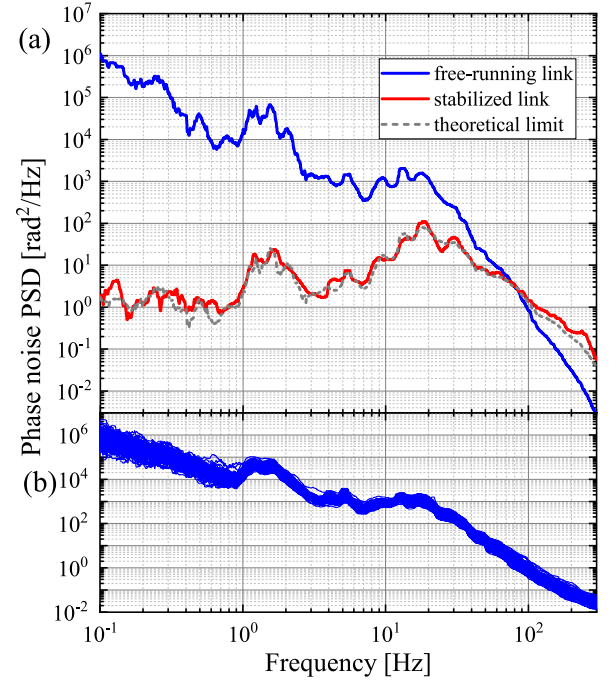


Fig. 2. (a) Measured phase noise PSDs for the running 260 km urban fiber link (blue curve), the stabilized fiber link (red curve) and the theoretical prediction (gray curve) (b) Measured phase noise PSDs of the 260 km free-running fiber for a whole day. The average phase jitter integrated from 0.1 Hz to 100 kHz of this free-running 260 km fiber link is about 240 rad.

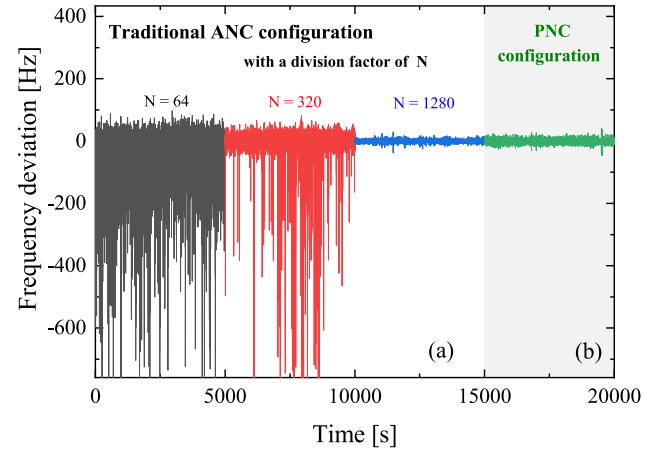


Fig. 3. Temporal behavior of the end-to-end frequency values of the stabilized 260 km urban fiber link respectively with the traditional ANC configuration (a) and the PNC configuration (b).

3.3. Cycle-slip free optical frequency transfer

To examine the phase noise rejection capability of the ANC and PNC techniques, we measured end-to-end frequencies in these two configurations, respectively. The ANC system we used is similar with our previous optical frequency transfer system [25]. In this configuration, the angular frequency of E_3 indicated in Eq. (4) is divided by a programmable frequency divider and sent to a phase comparator. The rule of thumb for the frequency dividing ratio is that the integrated phase jitter of the RF signal after frequency divider has to be much less than 1 rad. As shown in Fig. 3(a), the optical frequency transfer system could not perform properly when the frequency division ratio is 64. By increasing the ratio to 320, the system could be successfully

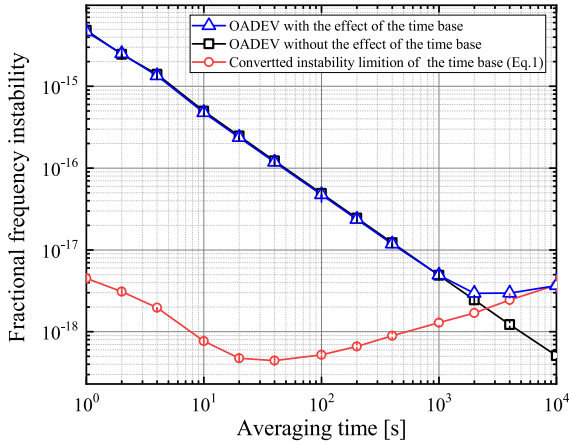


Fig. 4. Measured OADEV with (blue triangles) and without (black squares) the effect of the time base. The time base is from a quartz oscillator. To clearly illustrate the effect of the oscillator, the stability limited by the oscillator is also shown in red circles.

locked, but there still exist many occasional cycle-slips because the free-running phase noise of this fiber link is not stationary as shown in Fig. 2(b). The system could operate properly when the division factor is increased to 1280. Fig. 3(b) shows the temporal behavior of the end-to-end frequencies in the passively stabilized fiber link, demonstrating that the PNC configuration could run properly without the clear indication of the cycle-slips. As can be seen from the above experimental results, the ANC system needs a reasonable design for the frequency dividing ratio and the loop parameters, while the PNC-based optical frequency transfer system is more efficient, simple, and robust employed in a noisy fiber link.

3.4. Demonstration of the time base free optical frequency transfer

To illustrate the effect of the time base at the local site [12–14,22,23], before feeding the RF signal onto the AOM1 we mix it with a RF signal E_s with the angular frequency of ω_s . This measurement is based on the 150 km spooled fiber link. The time base from a quartz oscillator converting to the optical frequency shows a instability of 4.5×10^{-18} at 1 s and 2.9×10^{-18} at 10,000 s in terms of the OADEV as illustrated the red circle markers in Fig. 4. We can clearly see that, once E_s is employed in the system, the frequency of the fiber output light depends on the time base even when the PNC is properly working, as shown with the blue triangles in Fig. 4. The frequency instability is 4.9×10^{-15} at 1 s averaging time, and is limited to the 10^{-18} level at the long averaging time due to the effect of the time base. Once E_s is removed, the relative frequency instability is 4.9×10^{-15} at 1 s averaging time and scales down to 5.1×10^{-19} at 10,000 s averaging time (black squares), independent of the time base at the local site. We can conclude that with the implementation of the proposed technique, the effect of the time base can be effectively avoided.

3.5. Optical frequency transfer instability characterization

Complementary to the frequency-domain feature, the time-domain characterization of the corresponding fractional frequency instability in the 260 km noisy urban fiber link is shown in Fig. 5. The fractional frequency instability of the end-to-end frequency measurements are expressed in terms of the OADEV (black squares) and modified Allan deviation (MDEV, red circles). With the implementation of the PNC technique on the 260 km urban fiber link, we demonstrate a instability of 4.9×10^{-14} at 1 s integration time and scales down to 4.7×10^{-18} at the integration time of 10,000 s for the OADEV with a slope of τ^{-1} , while the MDEV for the same frequency data decreases to 5.1×10^{-20} at 10,000

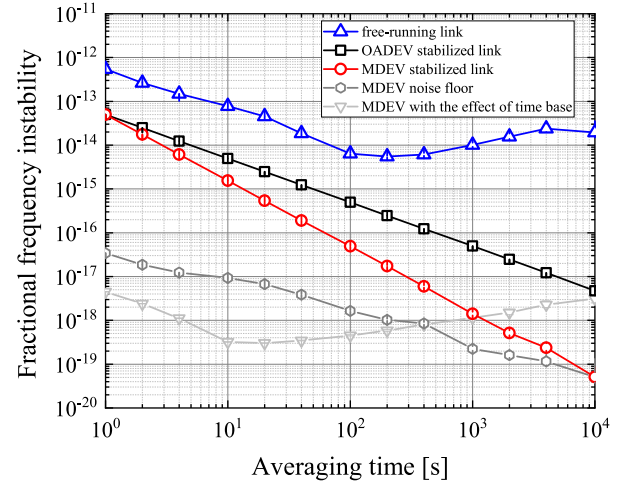


Fig. 5. Measured fractional frequency instability of the stabilized 260 km urban fiber link in terms of the OADEV (black squares) and the MDEV (red circles) with a Π -type frequency counter. Measured fractional frequency instability of the free-running fiber link (blue triangles) and noise floor of this fiber-based interferometer (gray hexagons) are also shown.

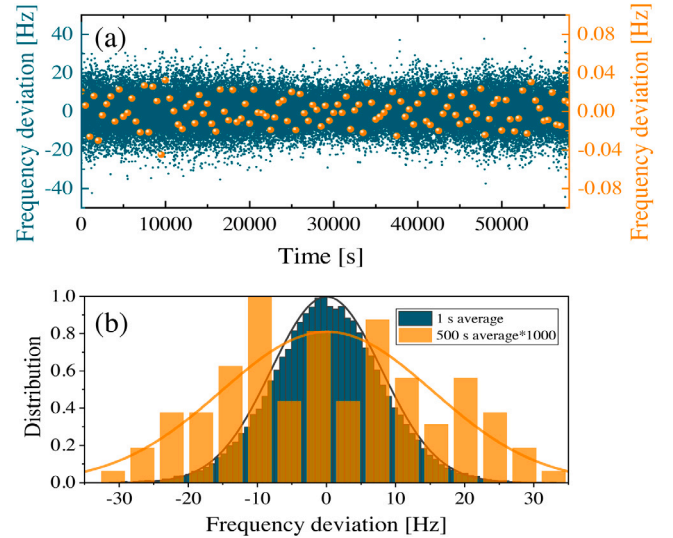


Fig. 6. (a) Frequency comparison between sent and transferred optical frequency through the 260 km urban fiber. The 58,000 data points were taken with a dead-time free Π -type frequency counters with a gate time of 1 s (dark cyan points, left axis). The unweighted arithmetic means for all cycle-slip free 500 intervals have been calculated, resulting in 116 data points (orange dots, right axis). (b) Histograms and Gaussian fits for the obtained frequency values. A relative difference between sent and transferred frequency of $(0.41 \pm 4.7) \times 10^{-18}$ is calculated.

s as a slope of $\tau^{-3/2}$. The results indicate that the white phase noise is dominated in the phase-compensated fiber link. In our experiment, we observe that the instability of optical frequency transfer is improved by three orders of magnitude at the integration time of 10,000 s by activating the PNC setup.

3.6. Frequency transfer accuracy analysis

To reveal the existing frequency offset of the transmitted light which may not be disclosed in the instability analysis, we further perform the evaluation of the optical frequency transfer accuracy. Fig. 6(a) shows the frequency deviation of the beat-note's data, recorded with a 1 s gate time and the total 58000 data points (left axis). By averaging the data every 500 s, resulting in 116 data points as shown in the right axis of

Fig. 6(a). The total 116 data points have an arithmetic mean of $78.9 \mu\text{Hz}$ (4.1×10^{-19}) and a standard deviation of 9.2 mHz (4.7×10^{-17}), which is a factor of 500 smaller than the OADEV at 1s as expected for this Π -type evaluation. Considering the long-term instability of frequency transfer as illustrated in Fig. 5, we conservatively estimate the accuracy of the transmitted optical signal as shown in the last data point of the OADEV, resulting in a relative frequency accuracy of 4.7×10^{-18} . We can conclude that there is no systematic frequency offset between the sent and transferred frequencies at a level of a few 10^{-18} . The Histograms and Gaussian fits for the frequency deviations are respectively illustrated in Fig. 6(b), the slight deviation from the Gaussian shape arises from temperature drifts in the asymmetric fiber parts, which are not included in the correction system. To suppress this influence, a more sophisticated package and advanced temperature control are demanded.

4. Discussion and conclusion

In summary, we for the first time demonstrated coherence transfer of an optical frequency through a 260 km-long urban fiber with the PNC technique without the effect of the time-base at the local site. Neither precise phase detector with the large dynamic range nor phase locked loop are longer needed in this PNC-based optical frequency transfer system, lowering the probability of cycle-slips for a noisy fiber link. Moreover, getting rid of the time base effect maintains the long-term instability of the optical frequency reference. After being compensated, the relative frequency transfer instability is 4.9×10^{-14} at 1 s averaging time and 5.1×10^{-20} at 10,000 s averaging time in terms of the MDEV. The frequency uncertainty of the light after transferring through the fiber achieves $(0.41 \pm 4.7) \times 10^{-18}$. Although the residual phase noise with the PNC technique is 7 times higher than that in the conventional ANC system, the phase noise could be suppressed by dividing the whole link into several sub-sections with laser repeater stations, a method which is also applied in ANC system [16,24,38]. Although here we only demonstrate that the time base effect can be effectively avoided in the PNC-based optical frequency transfer, it can also be used for the ANC-based scheme by slightly modifying the configuration of the setup. The proposed technique demonstrated here opens a way to a broad distribution of an ultrastable frequency reference with high coherence without any effects coming from the RF reference and enables a wide range of applications beyond metrology over fiber networks.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

The authors are unable or have chosen not to specify which data has been used.

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